



---

## Security Metrics and Executive Reporting

### Overview

This project demonstrates how to measure and report key Security Operations Center (SOC) performance metrics using **Elastic Security**, **Google Sheets**, and **Google Docs**. It focuses on actionable insights for improving detection and response capabilities.

---

### Tools Used

- Elastic Security (SIEM & dashboards)
- Google Sheets (metrics analysis)
- Google Docs (executive reporting)

---

### Key Metrics Dashboard

A dashboard was designed in Elastic Security to track:

- **Mean Time to Detect (MTTD):** 2 hours
- **Mean Time to Respond (MTTR):** 4 hours
- **False Positive Rate:** 12%

### Purpose

These metrics provide visibility into SOC efficiency:

- Lower MTTD → faster threat detection
- Lower MTTR → faster incident containment
- Lower false positives → reduced analyst fatigue

---

### Executive Summary (150 words)

The Security Operations Center (SOC) performance was evaluated using key operational metrics, including Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), and false positive rate. Current findings show an MTTD of 2 hours and an MTTR of 4 hours, indicating moderate detection and response efficiency. However, a false positive rate of 12% highlights opportunities for improving alert accuracy and reducing analyst workload.

To enhance SOC performance, it is recommended to fine-tune detection rules within Elastic Security, implement threat intelligence enrichment, and automate repetitive triage tasks.

Additionally, improving correlation rules and leveraging machine learning capabilities can help reduce noise and improve detection precision.



Overall, while the SOC demonstrates solid foundational capabilities, targeted optimizations in detection engineering and response automation will significantly improve operational efficiency, reduce risk exposure, and strengthen the organization's overall security posture.

---

## **Metrics Analysis (Google Sheets)**

### **Dwell Time Analysis (50 words)**

The mock incident analysis revealed a dwell time of approximately 6 hours between initial compromise and detection. This indicates delayed visibility into attacker activity. Improving log correlation, endpoint visibility, and real-time alerting can significantly reduce dwell time and limit potential damage during future incidents.

---

## **Recommendations**

- Improve detection rule tuning to reduce false positives
  - Implement automated response workflows (SOAR)
  - Enhance endpoint and network visibility
  - Use threat intelligence feeds for better context
  - Continuously monitor and refine SOC metrics
- 

## **Conclusion**

Tracking and analyzing SOC metrics such as MTTD, MTTR, and dwell time enables data-driven decision-making. By optimizing these areas, organizations can significantly improve their incident detection and response capabilities.

---